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Agentic AI

The Real-World Business Problem

In the modern Supply chain, e-commerce and retail businesses often suffer from inventory stagnation. While "out-of-stock" issues get immediate attention, the opposite, lack of SKU engagement or declining order frequency, often goes unnoticed until the end of a quarter. This leads to increased holding costs, forced markdowns, and wasted marketing spend on products that are no longer resonating with the target audience. The goal of this agent is to help corporations in the supply chain sector to target the least engaged products or SKUs and act as a “Business Intelligence” agent to uncover why these products aren't purchased.

Target Users

- Inventory Managers: Who need to decide whether to liquidate, discount, or restock.
- Marketing Analyst/ Data Analyst: Who need to know which specific SKUs require a "boost" in visibility.
- Small to Mid-sized E-commerce Owners: Who lack the time to manually audit thousands of SKU performance metrics daily.

Why an Agentic Solution?

Traditional software provides static dashboards (e.g., Tableau & Power BI), leaving the human to interpret data and decide on the workflow. An agentic solution is superior because:

- Autonomous Monitoring: It doesn't wait for a human to log in; it proactively identifies "at-risk" SKUs.
- Dynamic Workflow: Based on the reason for the decline (e.g., a new competitor price vs. a bad recent review), the agent can pivot its strategy (e.g., drafting a promotion vs. suggesting a product description update).
- End-to-End Execution: It moves from data analysis(ETL from the cloud) to conducting root cause analysis as to why engagement is low.

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2. Use Cases & Scenarios

Scenario A: The "Slow-Moving" Alert

- User Goal: A Warehouse Manager wants to identify items that haven't moved in 30 days and take action.
- Agent Tasks: 1. Data Fetcher queries the inventory database for SKUs with zero sales in the last 30 days.
  2. Market Analyst checks external competitor pricing for those specific items.
  3. Planner suggests a 15% discount or a "Bundle" strategy.

Scenario B: The "Negative Sentiment" Pivot

- User Goal: A Brand Manager notices a high-engagement SKU has suddenly stopped selling.
- Agent Tasks:
  1. Sentiment Agent scrapes recent customer reviews and social mentions.
  2. Critic Agent identifies a recurring complaint (e.g., "size runs small").

3. Summarizer drafts an updated product description and a customer service template to address the issue.

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3. Initial System Design

I will implement a Multi-Agent Orchestration

Preliminary Agents & Roles

- The Orchestrator (Planner): Decomposes the user's request and assigns tasks to specialized agents.
- Data Retrieval Agent: Interfaces with SQL/CSV files to pull SKU velocity, stock levels, and historical sales.
- Market Intelligence Agent: Uses web search tools to check competitor prices and current trends.
- ***Optional***Content/Copywriter Agent: Generates promotional emails, Slack alerts, or updated Shopify descriptions.
- The Critic: Reviews the proposed actions for budget compliance and logical consistency before presenting to the user.

Expected Tools & Data Sources

- Internal Data: Mock CSV/JSON datasets representing inventory levels and sales logs.
- Search Tools: Google Search API for competitor benchmarking.
- Communication: Slack API or email (SMTP) for delivering the final recommendations.
- LLM: gemini flash lite 2.5

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4. Feasibility & Risks

Risk	Mitigation Strategy
Hallucination	Use RAG (Retrieval-Augmented Generation) to ensure the agent only discusses SKUs present in the provided dataset.
Tool Failure	Implement robust error handling; if the "Search Tool" fails, the agent should proceed with internal data only.
Long Context	Use a "Summarizer" agent to condense historical sales data before passing it to the Planner.

Privacy	Use anonymized SKU IDs and mock customer data to ensure data privacy.

Timeline for Execution

Milestone	**Deliverable**	**Deadline**
Milestone 0	**Project Proposal** (Initial design and problem definition)	**Feb 04**
Milestone I	**Single-Agent/Front-End** (Build simple web/mobile interface)	**Mar 10**
Milestone II	**Multi-Agent Workflow** (Implement the full agentic interaction)	**Mar 24**
Milestone III	**Final Evaluation** (Comprehensive system assessment)	**Mar 31**